

SECTION 5 – POWER REQUIREMENTS

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5-1 CRITICAL POWER REQUIREMENTS

The critical power requirements for Signa MR/i and CV/i are unchanged from the requirements for Signa Horizon LX and CVMR. However, the upgrade modifies the existing system PDU to provide an appropriate circuit for the Water Chiller for Gradient Coil Cooling.

Note

The requirements for facility input power to the Signa MR/i system have been revised in *Direction 2223170 Signa MR/i & CV/i Pre-Installation*, Rev 2 SECTION 5 – POWER REQUIREMENTS. For convenience of reference the following table contains the revised requirements. For details and additional information refer to *Direction 2223170 Signa MR/i & CV/i Pre-Installation*, Rev 2.

TABLE 5-1
FACILITY INPUT POWER REQUIREMENTS

PARAMETER	REQUIREMENTS																											
CONFIGURATION	1.5T SYSTEM WITH COMPACT PDU CABINET: – 3 phase WYE with Neutral and Ground (5 wire system) 1.0T SYSTEM WITH RF/PDU CABINET: – 3 phase DELTA and Ground (4 wire system) Recommend Grounded Delta configuration – If 3 phase WYE with Neutral and Ground (5 wire system) input used then Neutral must be terminated inside the Main Disconnect Control and not brought to the RF/PDU Cabinet																											
INPUT VOLTAGE	480 Vrms 380/400/415 Vrms Note: A second circuit breaker is required for Shield/Cryo Cooler Cabinet.																											
FREQUENCY	50 ± 0.5 Hz or 60 ± 0.5 Hz																											
MINIMUM FACILITY TRANSFORMER REQUIRED FOR SINGLE ROOM APPLICATION	The facility transformer size to be determined based on the site layout, the load regulation, continuous sustained power, and peak instantaneous power requirements.																											
ALLOWABLE INPUT VOLTAGES/CURRENT DEMAND FOR SYSTEM INCLUDING SHIELD/CRYO COOLER See Note 1	<table><tr><th rowspan="2">NOMINAL VOLTAGE (Vrms)</th><th rowspan="2">NORMAL RANGE ±10 PERCENT</th><th colspan="2">CURRENT (AMPS)</th><th rowspan="2">^C RECOMMENDED MINIMUM STANDARD OVERCURRENT PROTECTION FOR ROOM FEED DISCONNECT</th></tr><tr><th>^A MAXIMUM MOMENTARY</th><th>^B CONTINUOUS</th></tr><tr><td>380</td><td>342–418</td><td>99</td><td>78</td><td>100 Amps</td></tr><tr><td>400</td><td>360–440</td><td>95</td><td>74</td><td>100 Amps</td></tr><tr><td>415</td><td>374–456</td><td>91</td><td>71</td><td>90 Amps</td></tr><tr><td>480</td><td>432–528</td><td>79</td><td>62</td><td>80 Amps</td></tr></table> NOTE: A Calculated at 59 KVA (50 KVA system Peak Instantaneous Power Demand + 9 KVA for Shield/Cryo Cooler Cabinet continuous operation). B Calculated at 46 KVA (37 KVA system Peak Instantaneous Power Demand + 9 KVA for Shield/Cryo Cooler Cabinet continuous operation). C Overcurrent protection sized for 125% continuous current per National Electrical Code NEC 1999 Article 210–20 or NEC 1996 Article 220–3.	NOMINAL VOLTAGE (Vrms)	NORMAL RANGE ±10 PERCENT	CURRENT (AMPS)		^C RECOMMENDED MINIMUM STANDARD OVERCURRENT PROTECTION FOR ROOM FEED DISCONNECT	^A MAXIMUM MOMENTARY	^B CONTINUOUS	380	342–418	99	78	100 Amps	400	360–440	95	74	100 Amps	415	374–456	91	71	90 Amps	480	432–528	79	62	80 Amps
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PEAK INSTANTANEOUS POWER DEMAND <5 SECONDS See Notes 1 and 2	For Base, HiSpeed, or EchoSpeed system configurations with SGD: 59 KVA including 9KVA (continuous operation) for Shield/Cryo Cooler Cabinet																											
CONTINUOUS SUSTAINED POWER (> 5 seconds) See Note 1	For Base, HiSpeed, or EchoSpeed system configurations with SGD: 46 KVA including 9KVA (continuous operation) for Shield/Cryo Cooler Cabinet																											
AVERAGE POWER DEMAND (Based On Typical 30 Minutes Per Hour Patient Scan Time)	For Base, HiSpeed, or EchoSpeed system configurations with SGD: 27 KVA at 0.9 lagging Power Factor including 9KVA (continuous operation) for Shield/Cryo Cooler Cabinet																											
SYSTEM IDLE QUIESCENT	For Base, HiSpeed, or EchoSpeed system configurations with SGD: 18 KVA at 0.9 lagging Power Factor including 9KVA (continuous operation) for Shield/Cryo Cooler Cabinet																											
TRANSIENTS	Maximum transient voltage above nominal waveshape not to exceed 200 V for the nominal voltages 380, 400, 415, or 480 Vrms.																											
PHASE IMBALANCE	Difference between the highest phase line–to–line voltage and the lowest phase line–to–line voltage must not exceed 2%.																											
LOAD REGULATION AT LINE FREQUENCY	Wires to be sized such that the line voltage drop from power source to PDU is less than 2% of the nominal voltage for rated load of system and with total power regulation for source power 4%, i.e. includes all feeders and transformers to utility.																											
GROUNDING	• Main facility ground wire to be minimum 1/0 AWG copper or same size as power feeder wire, which ever is larger. • Main facility ground wire to be insulated. • Ground impedance to earth at power source to be 2 ohms or less. • Main facility ground wire to be bonded at every distribution box in an approved grounding block.																											
Note 1 Breaker feeding system to be sized for Continuous Sustained Power. 2 Feeders must be sized for Peak Instantaneous Power Demand.																												